Our Proposed AI Education Model

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1. Purpose

Our goal was to create a model that synthesizes 20 academic sources on artificial intelligence in education, with emphasis on learning outcomes, self-regulated learning, assessment, ethics, and employability. It includes: (1) a comparative State of the Art table, (2) a critical analysis of trends, strengths, and gaps, and (3) a proposed model (LEAP-Career+ with guardrails) supported by the evidence base and summarized in a diagram.

2. Comparative State of the Art Table

Author / Year	Context	Method	Model / Proposal	Key Findings	Limitations	Contribution to My Model
Bond et al. (2024)	Higher Education (global)	Meta systematic review (review of reviews; 66 studies)	AI in Higher Education research landscape + ethics agenda	AI in HE is expanding rapidly; calls for stronger ethics, collaboration, and methodological rigour; governance is becoming central.	Synthesizes secondary studies only; coverage limited to reviews through mid-2023; English-language bias.	Use as evidence for an Ethics/Guardrails pillar and for research-based implementation (evaluation + rigor).
Limna et al. (2022)	Education (cross-sector)	Systematic review (multi-database)	AI-in-education adoption themes (personalization, assessment, analytics)	AI supports personalization and feedback but requires teacher capacity, data governance, and equity-aware design.	Broad scope; limited consistent empirical effect sizes; some included studies pre-GenAI.	Use to justify baseline AI literacy + institutional readiness components (training + infrastructure).
Kestin et al. (2025)	Higher Education (physics course)	Randomized Controlled Trial (crossover; N≈194)	AI-powered tutor vs in-class active learning	AI tutoring condition produced higher learning gains than active-learning classes under identical content; student perceptions vary by pace/experience.	Single course/institution; short-term outcomes; AI tutor quality depends on design and guardrails.	Use as core evidence for an Adaptive AI Tutoring pillar + emphasize need for design, pacing, and safeguards.
Liu et al. (2023)	Higher Education (EFL reading)	Controlled experiment (4 weeks)	ChatGPT-assisted reading & learning support	ChatGPT-assisted group improved English reading performance and engagement when studying original novels.	Short duration; single context; risk of over-reliance on AI support.	Use to support targeted language-learning use cases and to include 'AI coaching' with transparency rules.
Chen et al. (2022)	Higher Education	Design + evaluation (two studies)	AI student assistant / chatbot for course help	Chatbot-style assistants can increase immediacy of help, Q&A coverage, and student satisfaction when integrated with course workflows.	Generalization depends on domain; requires content curation and maintenance; privacy concerns.	Use to justify AI mentorship/assistant role and workflow integration (office hours, FAQ, resource routing).
Sajja et al. (2024)	Higher Education	Design/architecture + evaluation (intelligent assistant)	Personalized & adaptive learning assistant	Proposes an intelligent assistant that adapts resources/feedback to learner profile and context; highlights integration with analytics.	Implementation complexity; reliance on high-quality data and careful evaluation.	Use to design personalization loop (learner model → recommendation → feedback → update).
Weng et al. (2024)	Higher Education	Scoping review	GenAI implications for assessment & learning outcomes	Assessment practices are being reshaped (authentic assessment,	Scoping (not effect-size); many early/short studies; heterogeneous	Use to define assessment redesign and integrity policy

				process evidence, AI disclosure); research trends show rapid growth and unresolved integrity issues.	outcomes.	inside Policy & Practice Alignment pillar.
Lam et al. (2024)	Higher Education	Empirical study (writing task w/ ChatGPT)	ChatGPT as AI writing assistant	ChatGPT support can improve drafting fluency/idea generation but may reduce ownership without scaffolding and citation norms.	Task-specific; quality depends on prompting literacy; academic integrity risk.	Use to set AI writing guidelines (process logs, citation, reflection) and AI literacy outcomes.
Anik et al. (2023)	Higher Education (IT)	Mixed-methods / case study	ChatGPT in IT education ecosystem	Maps benefits (debugging help, explanations) and risks (hallucinations, shallow learning) in IT education; proposes integration patterns.	Early evidence; context-limited; outcomes often self-reported.	Use to include discipline-specific 'AI use protocols' and verification habits (testing, source checking).
Söderström et al. (2024)	Higher Education	Survey/interviews	Teachers' and students' views on ChatGPT	Stakeholders value efficiency and support but worry about plagiarism, misinformation, and assessment fairness; need clear guidance.	Perception-focused; regional sample; attitudes may shift over time.	Use to include change-management: staff training, student orientation, and shared policy language.
Medová et al. (2025)	K-12/Community STEM	Design-based / co-design study	Interdisciplinary MathCityMap trail (co-designed learning)	Co-design supports engagement and place-based problem solving; identifies constraints (time, coordination) and opportunities (interdisciplinary learning).	Not GenAI-centered; context-specific; outcomes depend on facilitation.	Use to strengthen 'authentic tasks + project portfolios' for employability evidence, even beyond GenAI.
Deng et al. (2024)	Higher Education	Systematic review (Computers & Education)	ChatGPT effects on learning	Synthesizes evidence that ChatGPT can support learning, especially feedback and explanation, but emphasizes variability and the need for pedagogical scaffolding.	Fast-evolving field; mixed study quality; limited longitudinal evidence.	Use to frame when ChatGPT helps (scaffolded) and to justify evaluation KPIs and guardrails.
Cipriano & Bauyot (2026)	Higher Education	Systematic review	Educational assistants (AI) overview	Finds AI assistants can aid support services and learning guidance; highlights governance, accessibility, and	Definitions vary; limited RCT evidence; reporting gaps.	Use to formalize roles: AI assistant + human mentor, plus accessibility and oversight requirements.

				human oversight as recurring requirements.		
Chang et al. (2023)	Education (cross-sector)	Design principles paper	AI chatbot supporting self-regulated learning (SRL)	Design principles: goal setting, timely feedback, and personalization; SRL support is strengthened when AI prompts reflection and planning.	Conceptual emphasis; empirical validation may be limited by context.	Use to structure SRL loop in model (goals → plan → practice → feedback → reflect).
Song & Song (2023)	Higher Education (EFL writing)	Quasi-experimental / classroom study	ChatGPT for AI-assisted language learning	ChatGPT-assisted learners improved academic writing skills and motivation; benefits linked to immediate feedback and idea support.	Short-term; potential over-dependence; quality/ethics concerns.	Use to justify adaptive feedback + motivation supports, paired with ethics and source transparency.
Schön et al. (2022)	Higher Education	Theory/review	How AI assistants are changing HE	Argues AI assistants shift teaching roles toward facilitation; stresses institutional strategy, ethics, and workforce alignment.	Not an experiment; outcomes depend on implementation quality.	Use for change-management and role redesign (faculty as coach; AI as support).
Jin et al. (2023)	Online learning (Higher Ed)	Systematic review / framework	AI chatbot to support self-regulated learning online	Shows SRL outcomes improve when chatbots provide metacognitive prompts, reminders, and adaptive feedback in online courses.	Heterogeneous measures; many small pilots; limited long-term follow-up.	Use to operationalize SRL features (nudges, dashboards, reflection prompts) and link to retention KPIs.
Park & Ahn (2024)	Higher Education	Focus groups + participatory design (students + stakeholders)	Opportunities/risks of ChatGPT + design implications	Identifies use cases (brainstorming, explanations) and harms (hallucinations, academic integrity, reduced deep learning); proposes design implications and stakeholder needs.	Qualitative; one-semester context; may not generalize across disciplines.	Use to incorporate stakeholder-centered guardrails and UX features (verification prompts, disclosure, citation support).
Wang & Fan (2025)	Education (multi-study synthesis)	Meta-analysis	Effect of ChatGPT on learning performance, perception, higher-order thinking	Overall positive effects on learning performance and perceptions, with moderated impacts on higher-order thinking depending on scaffolding and task design.	Publication bias risk; study heterogeneity; many short interventions.	Use as evidence base for expected learning gains and for focusing on higher-order thinking supports.
Bastani et al. (2025)	High school mathematics	Randomized field experiment (RCT)	Generative AI with/without guardrails	Shows that unrestricted GenAI can harm	Domain-specific; focuses on one subject;	Use to justify a Guardrails-by-Design

				learning outcomes; guardrails and guided use are crucial for benefit.	outcomes sensitive to enforcement of guardrails.	requirement across all pillars (verification, limits, teacher controls).

3. Critical Analysis

3.1 Commonalities

Across the corpus, AI is consistently framed as a lever for personalization and timely feedback, learner support at scale (e.g., chatbots and AI student assistants), and skills development aligned with employability. Multiple syntheses emphasize self-regulated learning (SRL) supports goal setting, reflection prompts, and adaptive feedback as mechanisms that can translate AI features into durable learning outcomes (Chang et al., 2023; Jin et al., 2023). At the same time, ethics and governance appear as a recurring requirement: meta-level work calls for stronger methodological rigour and explicit ethics practices (Bond et al., 2024), and field evidence shows that unguarded use can backfire (Bastani et al., 2025).

3.2 Differences

The sources differ substantially in their evidence type and level of maturity. Several are primary empirical studies (e.g., RCTs and classroom experiments) that report measurable learning effects (Kestin et al., 2025; Song & Song, 2023), while others are synthesis studies (systematic reviews, scoping reviews, meta-analyses) that aggregate early evidence and highlight heterogeneity (Deng et al., 2024; Weng et al., 2024; Wang & Fan, 2025). Another split is orientation: some papers are pedagogical/learning-science driven (SRL design principles, assessment redesign), others are stakeholder or ecosystem driven (teachers' and students' perceptions; IT education workflows), and others are policy/strategy oriented (institutional readiness, ethics, and governance).

3.3 Strengths in the Evidence Base

A key strength is that the folder includes multiple rigorous syntheses and at least one high-quality randomized study. The RCT comparing AI tutoring to active learning provides causal evidence that well-designed AI tutoring can produce learning gains (Kestin et al., 2025). Meta-analytic evidence also suggests overall positive effects on performance and learning perceptions, especially when tasks are scaffolded (Wang & Fan, 2025). Design-focused contributions add actionable guidance: SRL-oriented chatbot principles (goal setting, feedback, personalization) provide a bridge from 'AI feature' to 'learning mechanism' (Chang et al., 2023). Finally, stakeholder studies identify adoption realities and concerns about integrity, accuracy,

and fairness that must be addressed for sustainable implementation (Söderström et al., 2024; Park & Ahn, 2024).

3.4 Weaknesses and Gaps

Despite promising results, recurring gaps appear. Many interventions are short-term (weeks to one semester), and few measure longitudinal outcomes such as sustained skill growth or actual employability outcomes. Samples are often bounded to a single institution, course, or region, limiting generalizability. Across reviews, reporting quality is uneven: studies vary in how they describe the AI tool version, prompting approach, teacher involvement, and the presence/absence of safeguards, making replication difficult (Bond et al., 2024; Deng et al., 2024). Critically, negative or mixed effects are plausible: high school field evidence indicates that unrestricted GenAI can harm learning (Bastani et al., 2025), and qualitative work highlights hallucinations and shallow learning risks (Park & Ahn, 2024). Assessment remains a pressure point: educators are still converging on credible ways to assess learning in GenAI-rich environments while preserving integrity and fairness (Weng et al., 2024).

4. Proposed Model: LEAP-Career+

LEAP-Career+ is a practical framework for integrating AI into learning in ways that improve outcomes and employability while minimizing academic integrity, privacy, bias, and over-reliance risks. It is intended for higher education programs (but can be adapted to secondary education) and focuses on building student capability, not dependency.

4.1 Components (pillars)

Pillar 1: AI Literacy for All Disciplines. Students learn how LLMs work at a high level, how to prompt responsibly, how to verify outputs, and how to cite/attribute AI assistance. This pillar is motivated by readiness themes and stakeholder concerns (Limna et al., 2022; Söderström et al., 2024).

Pillar 2: Ethics and Guardrails. Implementation includes clear policies (disclosure, data privacy, bias awareness), tool constraints, and verification prompts. This is directly supported by calls for increased ethics and rigour (Bond et al., 2024) and evidence that unrestricted GenAI can harm learning (Bastani et al., 2025).

Pillar 3: Adaptive AI Tutoring. AI tutoring is deployed for practice, feedback, and

misconception repair, with pedagogical pacing and instructor oversight. Causal evidence shows potential learning gains when AI tutoring is carefully designed (Kestin et al., 2025).

Pillar 4: Career Integration and Mentorship. AI supports career exploration, interview preparation, and portfolio-building, but always paired with human mentorship and verification. This pillar is aligned with student assistant designs and IT education workflow analyses (Chen et al., 2022; Sakib et al., 2023).

Pillar 5: Policy and Practice Alignment. Curriculum, assessment, and labor market signals are aligned: assessments shift toward authentic tasks, process evidence, and reflective justification of AI use. Assessment redesign needs are highlighted in GenAI assessment reviews (Weng et al., 2024).

4.2 Transversal Competencies (foundation)

The model treats transversal competencies as the foundation: critical thinking (verification, skepticism), adaptability (tool changes), communication (prompting, explanation, documentation), and digital citizenship (ethics, privacy, respect). SRL research and design principles emphasize that these skills are strengthened when AI prompts goal setting, monitoring, feedback use, and reflection (Chang et al., 2023; Jin et al., 2023).

4.3 Evidence Mapping

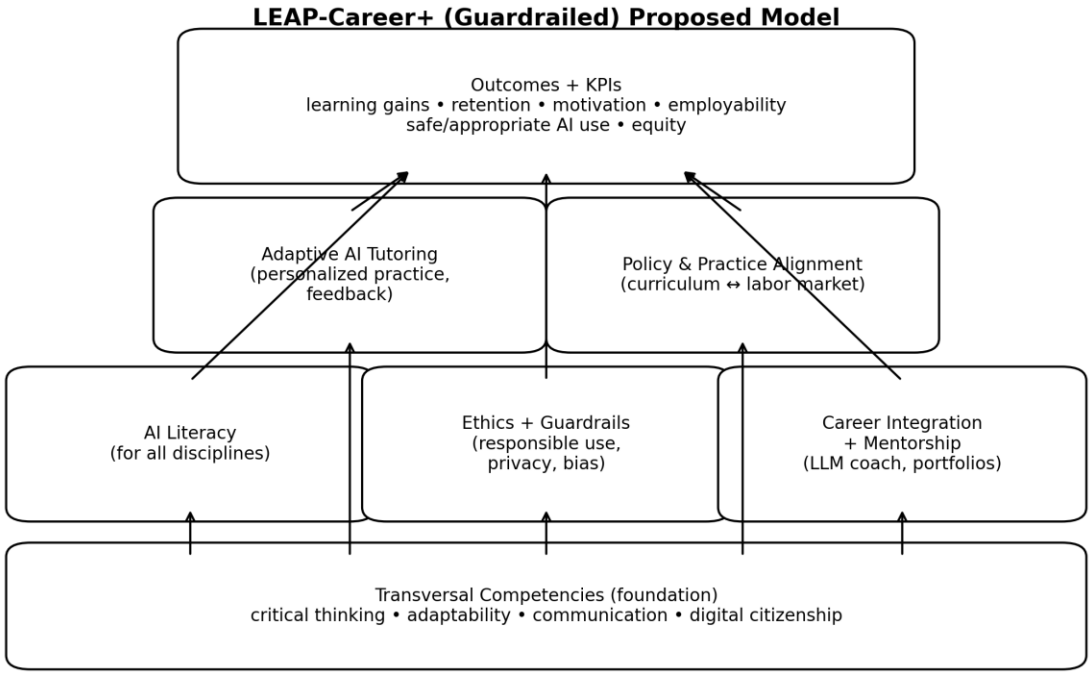
Empirical learning gains motivate the tutoring and language-learning use cases (Kestin et al., 2025). Syntheses motivate scope and boundary conditions: benefits are strongest when AI use is scaffolded and embedded in pedagogy (Deng et al., 2024; Wang & Fan, 2025). Guardrails and governance are treated as non-optional due to ethics-focused syntheses and harm evidence (Bond et al., 2024; Bastani et al., 2025; Park & Ahn, 2024). Finally, adoption depends on stakeholder trust and clear guidance (Söderström et al., 2024), so the model includes orientation, policy language, and instructor support.

4.4 KPIs

Recommended indicators include: (1) learning gains on course-aligned assessments, (2) retention and course completion, (3) SRL metrics (goal completion, reflection quality, help-seeking patterns), (4) academic integrity metrics (disclosure compliance, reduced misconduct incidents), (5) equity/access metrics (usage by subgroup, accommodation effectiveness), and (6)

employability outcomes (portfolio quality, internship placement, interview performance). KPIs should be tracked each term and used to tune guardrails and instructional design.

5. Model Diagram



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